

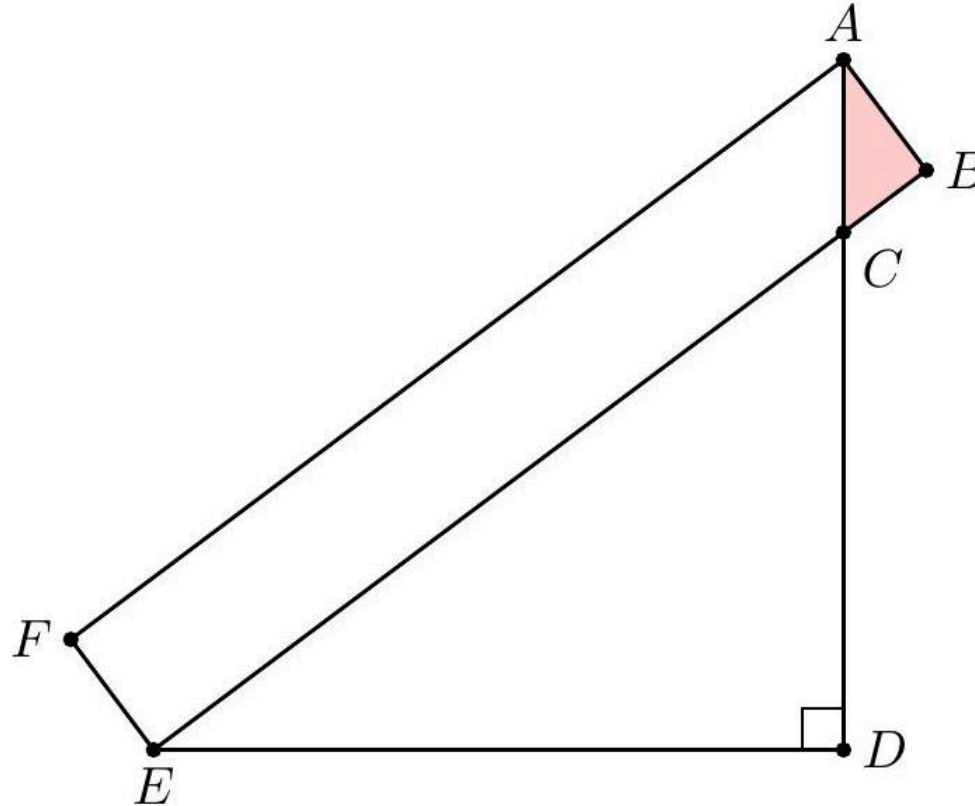
2025 AMC 10A Problem 15 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 15. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 15

Problem:

In the figure below, $ABEF$ is a rectangle, $\overline{AD} \perp \overline{DE}$, $AF = 7$, $AB = 1$, and $AD = 5$.



What is the area of $\triangle ABC$?

Answer Choices:

A. $\frac{3}{8}$

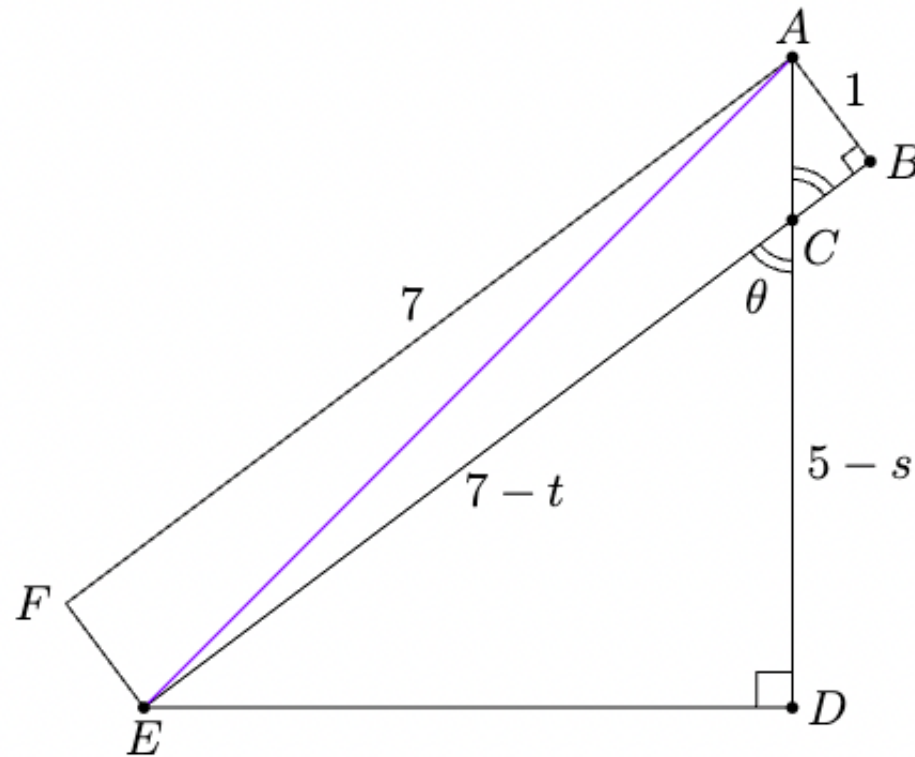
B. $\frac{4}{9}$

C. $\frac{1}{8}\sqrt{13}$

D. $\frac{7}{15}$

E. $\frac{1}{8}\sqrt{15}$

Solution:



Let the sides and points be as shown in the diagram. By the Pythagorean Theorem applied to the right triangles in the figure,

$$DE^2 = AE^2 - AD^2 = AB^2 + AF^2 - AD^2 = 50 - 25.25.$$

so $DE = 5$. Also, from the right triangle,

$$1 + t^2 = s^2 \quad (\text{Pythagorean Theorem}).$$

Using similar triangles ($\triangle ABC \sim \triangle EDC$),

$$\frac{AB}{DE} = \frac{AC}{CE} \Rightarrow \frac{1}{5} = \frac{s}{7-t} \Rightarrow 5s = 7-t.$$

Substituting $s^2 = 1 + t^2$,

$$25s^2 = 49 + t^2 - 14t.$$

Simplifying,

$$25t^2 - t^2 + 25 - 49 + 14t = 0 \Rightarrow 24t^2 + 14t - 24 = 0 \Rightarrow 12t^2 + 7t - 12 = 0.$$

Factoring,

$$12t^2 + 16t - 9t - 12 = 0 \Rightarrow (4t - 3)(3t + 4) = 0.$$

Hence,

$$t = \frac{3}{4}.$$

The area of $\triangle ABC$ is

$$[ABC] = \frac{1}{2}(\text{base})(\text{height}) = \frac{1}{2} \cdot 1 \cdot t = \frac{1}{2}t = \boxed{\text{(A)} \frac{3}{8}}$$

Alternate Solution:

From the figure in Solution 1, the triangles $\triangle ABC$ and $\triangle EDC$ are similar. Hence,

$$\triangle ABC \sim \triangle EDC.$$

By the Pythagorean Theorem, applied to the right triangle,

$$S^2 = 1 + t^2 \tag{1}$$

From the similarity of triangles,

$$\frac{5 - S}{7 - t} = \frac{t}{S} \tag{2}$$

Cross-multiplying,

$$7t - t^2 = 5S - S^2 \tag{3}$$

From (1) and (3),

$$7t + 1 = 5S.$$

Squaring both sides,

$$49t^2 + 1 + 14t = 25 + 25t^2.$$

Simplifying,

$$24t^2 + 14t - 24 = 0 \Rightarrow 12t^2 + 7t - 12 = 0.$$

Solving for t ,

$$t = \frac{-7 + \sqrt{49 + 576}}{2 \times 12} = \frac{-7 + 25}{24} = \frac{18}{24} = \frac{3}{4}.$$

Therefore, the area of $\triangle ABC$ is

$$[ABC] = \frac{1}{2} \times (\text{base}) \times (\text{height}) = \frac{1}{2} \times 1 \times t = \frac{1}{2}t = \boxed{(A) \frac{3}{8}}$$

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